

Failure modes & limitations

an honest look at where the heuristic genus breaks — and the guardrails that keep it safe

Purkinje cell

input: “a large GABAergic neuron of the cerebellar cortex”

picked **interneuron · CL:0000099**

should be **GABAergic neuron · CL:0000617**

The heuristic grabbed “GABAergic” — but Purkinje cells are projection neurons, not interneurons.

hepatic stellate cell

input: “a pericyte of the liver perisinusoidal space”

picked **cell · CL:0000000 (root)**

should be **pericyte · CL:0000669**

No groundable genus word in the name, so it falls back to the ontology root, not a specific parent.

Root causes

- **Genus derivation is a string heuristic**
hierarchically valid for only ~51% of terms (benchmark B2).
- **ELK checks logic, not biology**
a coherent-but-wrong genus still passes the reasoner.
- **The LLM only phrases**
it inherits upstream errors; it won't fix a wrong parent.
- **Markers need real expression data**
no matrix → specificity is prior-based, not measured.
- **Surface-marker → PRO grounding ≈ 83%**
gene-symbol / protein-name mismatches are missed.

Guardrails & roadmap

- **Curator-in-the-loop by design**
every dossier is a proposal — never auto-committed.
- **It shows its work**
grounded CURIEs + evidence → catch a wrong genus in seconds.
- **Duplicate detection worked here**
both cases were correctly flagged ALIGN to the existing term.
- **Confidence flags uncertainty**
both scored a moderate 0.55, not a false 1.00.
- **Roadmap: reasoner-derived genus**
take ELK's inferred parent + DRAGON-AI / CurateGPT RAG.